Research Scope, Timeline, and Authority

How to read the Formulation context corpus

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This folder explains the research. The `history/` folder holds source PDFs only. Emails and transcripts remain in their own folders. Do not treat this overview as a rewrite of those sources.

1 What this context tree is for

The `Formulation/context` tree is an archival reading guide for the SURA project on **algorithmic stability for post-hoc / optimization-selected policies**.

- `overview/` — this document: scope, timeline, authority, evolution, open problems.
- `history/` — source PDFs only (working survey + Wenbin problem setup).
- `emails/` — advisor email record that shaped direction.
- `transcripts/` — edited-verbatim research meeting record.
- `papers/` — reading library (priority / lab / background / tangential). Priority and pivot papers are in-scope for the evolution story; PDFs live here and are not duplicated into `history/`.

2 In scope / out of scope

2.1 In scope for this overview and `history/`

- `Algorithmic_Stability_Working_Document.pdf` (identical to Downloads `-5`/`-6`/`-7`).
- `Algorithmic_Stability_Wenbin_Problem_Setup.pdf` (Downloads `-8`; best available annotated draft / stand-in for the **missing** July 2 Overleaf; identity with that Overleaf is **unproven**).
- Advisor email packet (`emails/`).
- Meeting transcript packet (`transcripts/`): 27 Apr, 22 May, 29 Jun 2026.

2.2 Out of scope (not incorporated here)

- Lean operator catalog, certificates, verification pipelines under `Desktop/Research/Work`.
- Deep-research CSV/agent dumps outside this corpus.
- Earlier proposal drafts `Algorithmic_Stability-1` through `-4`, SURA tour PDFs, mentor–mentee forms, and other Downloads material not listed above.
- The `papers/` PDFs are *not duplicated into* `history/` (packaging). Priority/pivot papers **are** part of formulation history; they live in `papers/`.

Historical note. Excluding those materials is a *corpus boundary*, not a claim that they never mattered. If they are later brought into `history/`, this overview should be updated.

3 Authority hierarchy (read this first)

Authority is **layered by role**, not a single flat ranking. Swarm council (26 Jul 2026): do not crown H2 over Meeting 3 for everything; do not equate H2 with the July 2 Overleaf.

Layer	Controller	Controls
L0 Institutional	Woody > Wenbin > Nick	Mentorship / veto; does not auto-rank every PDF
L1 Research agenda	Meeting 3 (Woody)	Opt-induced **policy** stability; discontinuity; AI→optimizer; HT end product "trust this decision?"
L2 Written math object	July 2 Overleaf when recovered; **H2 = stand-in, identity unproven**	UQ $\hat{\theta}$ via constrained opt; goals (i)–(iii); Part I/II program; five `[Wenbin]:` comments high for *direction*
L3 Methods authorization	Wenbin 2 Jul email	Brainstorm methodologies for the L2 setup only; does not resurrect H1 as controlling framework
L4 Terminology / process	Nick-attested "post-hoc inference" (12 Jun); H2 W4 flexibility	Labels; permission to revise setting if more principal
L5 Historical / reference	H1 inventory; M1–M2; June 25 proposal; adaptive regret; quantization; CREAM backups	Non-controlling live inventory (not scrapbook); history only for frameworks

Conflict rules (council). (1) L1 constrains *class* (policy = opt solution). L2 fills *instance* (UQ / constrained $\hat{\theta}$). Woody endorsement of the UQ instance is **not yet in-corpus** — treat H2↔M3 as compatible on the surface, latent conflict still open.
(2) Methods must target L2 under L3. H1 supplies primitives/literature, not the problem definition.
(3) H2 Parts I/II are empty stubs — direction of formalization, not finished law.
(4) Never write "H2 = July 2 Overleaf."

4 Source inventory

ID	Document	Location	Status
H1	Stabilized Selection Algorithms (working document, 26 Jul 2026)	`history/Algorithmic_Stability_Working_Document.pdf`	Historical / Reference material (catalog still useful)
H2	Algorithmic Stability for Post-Hoc Selection (Wenbin comments + problem setup)	`history/Algorithmic_Stability_Wenbin_Problem_Setup.pdf`	Incomplete annotated draft (best available L2 stand-in; Parts I/II empty; Overleaf identity unproven)
E1	Advisor Email Record	`emails/Advisor_Email_Record_Research_Direction.{tex,pdf}`	Current (primary email archive)
T1	Research Meeting Transcripts	`transcripts/Research_Meeting_Transcripts_Synthesized.{tex,pdf}`	Current (edited verbatim; plan-status tags mark superseded ideas)

Triplicate note for H1. Downloads files `nmino___SURA___Algorithmic_Stability-5.pdf`, `-6.pdf`, and `-7.pdf` are byte-identical (same MD5). Only one copy is kept in `history/`.

4.1 Per-document classification headers

H1 — Working document (stabilized selection examples)

Document: Stabilized Selection Algorithms: Research working document
Date: 26 July 2026 (filename lineage: Downloads `-5`/`-6`/`-7`)
Type: Student working survey / example catalog
Status: Historical; catalog and taxonomies remain reference material

Purpose. Collect concrete examples of the form original selector → stabilized selector → formal certificate → utility/inference cost, as raw material for a possible composable stability-certification framework.

Relationship to later work. Supplies the notation $\hat{S} = A(D)$, $\tilde{S} = \tilde{A}(D;\xi)$, certificate types (stability / DP / post-selection CI), and primitive/mechanism/proof taxonomies. Presented in spirit at Meeting 3; later narrowed by Wenbin (H2) toward optimization-selected policies in UQ.

What remains valid. Example audit tables; Zrnic–Jordan noisy winner / screening / Stable LASSO writeups; DP selection primitives; synthesis that DP alone is not an inference bridge.

What changed later. Universal "composable framework for post-hoc selection algorithms" is narrowed in H2 to a specific UQ + constrained-optimization setting, with an explicit two-part research program.

Important ideas introduced. Unifying template; inclusion criteria; best-first proposed theorem (gap-aware noisy argmax/top-k → η-stability → CI correction).

Where those ideas appear. Notation and template echoed in H2's opening setup; Meeting 3 discussion of stabilized selection rules; email June 25 student proposal.

H2 — Wenbin problem setup (L2 stand-in; identity with July 2 Overleaf unproven)

Document: Algorithmic Stability for Post-Hoc Selection
Date: Compiled/annotated 26 July 2026 (Downloads `-8`). Same-lineage with Wenbin's 2 Jul Overleaf: **unproven**.
Type: Co-authored hybrid draft with inline Wenbin commentary; mixes student template, Meeting 3 remnants, and Wenbin UQ setup
Status: Incomplete annotated draft — best available L2 stand-in (not finished law)

Purpose. State the project as post-hoc inference after data-dependent selection, and narrow the general problem to an uncertainty-quantification setting where hyperparameter θ is chosen by constrained optimization on shared calibration data.

Relationship to later work. July 2 email authorizes methodologies for Wenbin's Overleaf setup. H2 is the best surviving candidate for that direction — **not** certified as the Overleaf text.

What remains valid. Five `[Wenbin]:` comments: intersection of post-hoc inference, algorithmic stability, and optimization; need to focus on one objective (UQ example); decompose into stability of constrained optimizations, then post-hoc inference; setting may be modified if a more principal version appears; strategy notes (data-driven policy / PTO; **no recalibration**; data-randomize; structured noise; inverse optimization; Tijana / winner's curse / algorithmic stability line).

What changed later / still incomplete. Sections 3.2–3.4 are stubs. Abstract TBA. Goal (iii) body still says "recalibrate" while Comment E prefers no recalibration — **unresolved mid-edit**.

Important ideas introduced. Validity assumption (1); optimization (2) defining $\hat{\theta}(D)$; goals (i)–(iii). **Dual text on (iii):** body asks how to recalibrate C; Comment E says no recalibration / data-randomize answers (iii). **Notation bug (do not sanitize):** introducer writes $C(X; D, \alpha)$ while Eq. (1) uses $C(X; D, \theta)$.

Where those ideas appear. Meeting 3 priorities (formal stability of optimization-induced policy; stabilization mechanism; trust the decision?); July 2 email authorizing methodology brainstorming.

E1 — Advisor email record

Document: Advisor Email Record — Research Direction
Date: Threads May 11 – July 3, 2026; packet compiled 26 July 2026
Type: Primary documentary email archive
Status: Current

Purpose. Preserve advisor-authored guidance and the student proposal that prompted reformulation.

Key preserved decisions. Reading priorities; Wenbin as technical guide; "post-hoc inference" terminology (Nick-attested, 12 Jun); June 25 primitive-operation proposal (unapproved by email); July 2 Wenbin Overleaf draft "aligns closer to what we intended" (text missing).

T1 — Meeting transcripts

Document: Research Meeting Transcripts (Edited Verbatim)
Date: Meetings 27 Apr, 22 May, 29 Jun 2026; packet compiled 26 July 2026
Type: Primary meeting archive
Status: Current (with explicit superseded plan tags)

Purpose. Retain substantive research discussion edited-verbatim; mark plan evolution so early proposals are not mistaken for the current agenda.

5 Chronological timeline

When	Event	Sources
27 Apr 2026	Meeting 1: three lab pillars (UQ for high-stakes decisions; grid resilience; human–AI collaboration). Nick to read and choose.	T1
11–14 May	Woody: read supplied papers; weekly meetings; PhD student forthcoming.	E1
18–20 May	Woody connects Nick with Wenbin; clarify research objectives from proposal. Papers named: inverse conformal risk control; conformalized decision risk assessment; Gen-DFL; hierarchical probabilistic conformal prediction.	E1
22 May 2026	Meeting 2: algorithmic pillar (CREDO/CREAM). Nick proposes adaptive certified regret frontier. Wenbin: Overleaf formulation; backups = sequential CREAM, model/solver error. Background: conformal prediction + OR optimization.	T1
8–11 Jun	Logistics confirm in-person meeting (Hamburg Hall / Teresa Heinz). Substantive June 11 notes not in this corpus.	E1
12 Jun	"Post-hoc inference" added to formal project description (per Nick's attestation of Wenbin's suggestion; Wenbin phrase not in email body).	E1
25 Jun	Nick emails primitive-operation / stabilized-selection approach for post-hoc selection/inference; requests meeting on Week 3 diagram.	E1
29 Jun 2026	Meeting 3: **current spoken agenda**. Stability of optimization-induced policies; discontinuity challenge; AI predictions → optimizer; end product as hypothesis test; priority papers PPI + algorithmic/post-selection stability (Angelopoulos & Jordan). Adaptive regret not reaffirmed.	T1
2 Jul 2026	Wenbin: Overleaf problem setup "aligns closer to what we intended"; Nick to brainstorm methodologies. Overleaf text **MISSING** from corpus.	E1; H2 (26 Jul) = unverified stand-in
26 Jul 2026	H1 working survey finalized; H2 annotated draft compiled same day (CreationDate later than H1). Corpus fold-	H1; H2; this file

6 Evolution of research questions

6.1 Idea chain (do not collapse)

Idea A — Quantization → risk-profile comparison. **First appearance:** before / at Meeting 2 (Nick).
Status: Redirected / no longer primary.
Why: Woody redirected toward pure algorithmic decision analysis, not engineering testing.
Remnant: Reappears only as a special case of "model error" in Wenbin's Meeting 2 backup ideas.

Idea B — Adaptive / certified bounded-regret frontier. **First appearance:** Meeting 2 (active plan); Overleaf formulation assigned.
Status: De facto not the active plan as of Meeting 3; no spoken "abandon forever"; revive unknown.
Why: Project pivoted to stability of optimization-induced policies and post-selection stability literature (Woody reframes; packet editorial marks superseded).
Historical note. Retain for intellectual history and any notation borrowed from CREAM during the Overleaf attempt (that Overleaf text is missing — see Missing evidence).

Idea C — Sequential CREAM; accommodate model/solver error. **First appearance:** Meeting 2 (Wenbin backups).
Status: Backup only; not adopted as Meeting 3 direction; revive unknown.
Status label: Not active (do not write bare "Abandoned" as if permanently killed).

Idea D — Stabilized selection algorithms + primitive catalog. **First appearance:** Nick email 25 Jun; literature survey shown at Meeting 3; H1 PDF written 26 Jul (same day as H2 compile — not an earlier artifact).
Status: Non-controlling live inventory / Historical reference (not scrapbook).
Evolution: Woody (M3) specializes object class to opt-induced policies; Wenbin (H2) further offers UQ hyperparameter-via-constrained-opt as working instance (Woody endorsement of that UQ instance not yet in-corpus).

Idea E — Stability of optimization-induced policies; trust the decision? **First appearance:** Meeting 3 (Woody), linking prior Woody–Wenbin–Ryan discussion.
Current form: L1 = M3 agenda; L2 working math = H2 UQ block ($C(X; D, \hat{\theta}(D))$ when $\hat{\theta}$ solves constrained opt) — with unresolved (iii)/Comment E clash (recalibrate vs no recalibration / data-randomize).
Status: Current (agenda + working direction).

6.2 Evolution (overlapping documents)

Three distinct earlier objects (do not collapse): (1) June 25 email proposal; (2) Meeting 3 example survey (possibly Downloads -2); (3) H1 PDF dated 26 Jul 2026. Shared pattern: general composable stability-certification for post-hoc selectors; template $\tilde{S} = \tilde{A}(D; \xi)$; certificates DP / (η, τ, ν)-stability / selective CIs.

↓

What changed: Woody (M3) restricts class to policies that are optimization solutions; Wenbin (H2) offers UQ hyperparameter-via-constrained-opt as working instance + Part I/II; method bias toward structured / data randomization (Comment E), while goal (iii) text still says recalibrate.

↓

Why it changed: Wenbin (H2): "The post-hoc inference problem is itself very general. We need to narrow…" Woody (Meeting 3): policy is solution to an optimization problem; define stability for that class first.

↓

Current understanding (L1 Meeting 3 + L2 H2 stand-in): study when optimization-selected $\hat{\theta}(D)$ preserves uncertainty-set validity; if not, quantify deviation; restore (1) preferably via data-randomize / structured noise (**not** settled recalibration of C — mid-edit clash). Keep work inside Woody's opt-induced policy class.

7 Evolution of the mathematical framework

7.1 Student unifying template (H1) — preserved

$$\hat{S} = A(D), \qquad \tilde{S} = \tilde{A}(D;\xi), \qquad \mathrm{Cert}(\tilde{A}) \le \eta \quad or \quad (\varepsilon,\delta){-}DP \quad or \quad \Pr\{\theta_{\tilde{S}} \in C_{\tilde{S}}(D)\} \ge 1-\alpha.$$

Pattern: primitive + mechanism + proof strategy ⇒ certificate at utility/inference cost.

7.2 Wenbin / working L2 setup (H2) — direction, not finished law

Primary inconsistency (keep visible): H2 introduces $C(X;D,\alpha)$ then uses $C(X;D,\theta)$ in (1). Below follows Eq. (1)'s θ form.

Uncertainty set $C(X;D,\theta)$ with validity for fixed θ:

$$\Pr(Y \in C(X;D,\theta)) \ge 1-\alpha.1$$

Data-dependent hyperparameter from constrained optimization:

$$\pi: \quad \min_{\theta} \hat{f}(y;\theta) \quad s.t. \quad \hat{g}(\theta) \le 0, 2$$

with $\hat{f}, \hat{g}$ depending on shared data D (filtration note in H2). Goals: (i) when does (1) still hold for $\hat{\theta}(D)$; (ii) miscoverage deviation if not; (iii) **as written** = recalibrate C; **Comment E override** = no recalibration, data-randomize answers (iii). Unresolved mid-edit.

Notation change (documented, not overwritten).

- Old (H1 / Meeting 3 slides): $\hat{S} = A(D)$, $\tilde{S} = \tilde{A}(D;\xi)$, confidence set for $\theta_{\tilde{S}}$.
- New (H2 UQ specialization): hyperparameter θ / $\hat{\theta}(D)$; uncertainty set $C(X;D,\theta)$; objective/constraint $\hat{f}, \hat{g}$.
- Reason: narrow from arbitrary selected object to optimization-chosen UQ hyperparameter (and, in Meeting 3 language, optimization-induced policy).

Historical note (Meeting 3). Woody asked Nick to define θ / $\tilde{s}$ in the slide notation; definitions were incomplete at the meeting. H2 is the later attempt to formalize; stubs in H2 §3.2–3.4 mean the formalization is still open.

8 Evolution of theorems and proofs

This corpus contains **example proofs from the literature** (H1) and **proposed / incomplete project theorems**, not completed original Lean proofs.

Claim	**O**
Zrnic–Jordan Thm. 2 style: (η, τ, ν)-stable selector $\Rightarrow$ corrected CI coverage	H1 §5 5.3 (lit er- a- tu
Proposed "best first theorem": gap-aware / heterogeneous noisy argmax/top-k $\to$ η-stability $\to$ CI correction	H1 §8
H2 goals (i)–(iii) for $C(X; D, \hat{\theta}(D))$	H2
Part I: stability of constrained optimizations under one-point data change	H2 W bi co m
Part II: post-hoc inference after stable selection	H2

9 Meeting record (extracts to later documents)

Full edited-verbatim text lives in `transcripts/`. This section only indexes advisor suggestions, decisions, directions, and open problems, with pointers.

9.1 Meeting 1 — 27 Apr 2026 (Woody, Nick)

- **Advisor suggestion:** Read publication list; choose among three pillars; ideally propose an unexplored follow-on.
- **Decision:** No concrete project title yet — problem space only.
- **Later:** Pillar choice $\to$ algorithmic (Meeting 2).

9.2 Meeting 2 — 22 May 2026 (Wenbin, Nick)

- **Decision (then):** Pursue adaptive regret frontier; write mathematical formulation in Overleaf using CREAM notation.
- **Advisor suggestions:** Background skim — conformal prediction; LP/stochastic programming. Backups — sequential CREAM; prediction/solver error.
- **Historical note:** Active plan at the time; **superseded** by Meeting 3.
- **Later documents:** E1 (June emails); T1 Meeting 3 plan-status table. Meeting 3 pivot supersedes this formulation path; H2 is a later formalization attempt (not the replacement agent by itself).

9.3 Meeting 3 — 29 Jun 2026 (Woody, Wenbin, Nick)

- **Decisions:** Current plan = formalize stability of optimization-induced policies; find stabilization for valid inference; applications with AI predictors + optimizer; end product as hypothesis testing ("should I trust this decision?").
- **Advisor suggestions:** Priority reading — prediction-powered inference; algorithmic / post-selection stability (Angelopoulos & Jordan). Share stabilized-selection example compilation by email.
- **Mathematical idea:** Core challenge = discontinuity / discrete decision boundaries under input perturbation (optimization structure).
- **Action items:** Formal problem setup; continue literature survey; discuss applications with Wenbin; pin Woody on Slack when ready for next meeting.
- **Later documents:** E1 July 2 Wenbin setup; H2 problem statement; H1 catalog.

10 Advisor feedback history

- **Woody (May):** Prioritize group papers; work with Wenbin on objectives.
- **Wenbin (12 Jun):** Nick attests Wenbin suggested adding "post-hoc inference" to the formal description (Wenbin's own sentence not in the email body).
- **Woody (29 Jun):** Align with long-discussed stability-of-optimization-policy idea; clarify notation; PPI + post-selection stability reading; discontinuity challenge.
- **Wenbin (2 Jul email):** Drafted a version of the Overleaf problem setup closer to intended problem; begin methodology brainstorming (Overleaf text missing from corpus).
- **Wenbin (H2 inline) — highest-authority *comments* in the surviving draft:**
  1. Problem sits at intersection of post-hoc inference, algorithmic stability, and optimization theory.
  2. Narrow general post-hoc inference by focusing on one objective (UQ example given).
  3. Decompose: (1) stability of constrained optimizations (one-point change; structural assumptions on $\hat{f}, \hat{g}$); (2) post-hoc inference on top.
  4. Setting may be modified if a more principal version appears.
  5. Strategy notes: data-driven policy (e.g. PTO); no recalibration vs data-randomize for validity; structured noise / inverse optimization; post-inference line (Tijana; winner's curse; algorithmic stability).

11 Literature integration

- **Lab algorithmic line (early reading):** CREDO/CREAM; inverse conformal risk control; conformalized decision risk assessment; Gen-DFL — directed Nick into the algorithmic pillar (Meetings 1–2; E1).
- **Pivot papers (Meeting 3):** Prediction-powered inference; post-selection inference via algorithmic stability (Zrnic & Jordan / Angelopoulos–Jordan line). These changed the active question from regret frontiers to stable selection / trustworthy AI+optimization decisions.

- **H1 catalog:** Large set of MAIN examples (noisy max, stability selection, DP top-k, Thresholdout, randomized selective inference, ...) plus baselines (Lee et al. LASSO SI; PoSI; winner inference without stabilization; sample splitting).
- **Library location:** `papers/` with `INDEX.md`; not duplicated in `history/`.

12 Operator / primitive catalog evolution

Present only in H1 (§8 taxonomies). Status: **Reference material**.

Primitives listed: noisy scalar release; noisy argmax; noisy top-k; noisy threshold; validation comparison for selective inference; DP validation/tuning; support/model selection; randomized optimization/ERM; subsample aggregation; private candidate selection; reusable holdout; iterative atom selection; best-arm selection.

Relation to current plan. Catalog remains raw material. H2 asks which primitives (especially structured noise tied to inverse optimization / constrained opt) can serve Part I–II; that mapping is still open.

13 Verification / discovery frameworks

Not present in this corpus. No verification outputs, Lean certificates, or discovery pipeline docs were included in the scoped materials. If added later, place sources under `history/` (or a dedicated folder) and extend this overview.

14 Open problems

Question	Status
Formal definition of stability for a policy induced by an optimization problem (Meeting 3; H2 Part I)	Still open
Stabilization mechanism enabling valid optimization-based post-hoc inference (Meeting 3; H2 (iii))	Still open
When does UQ validity (1) survive plugging in $\hat{\theta}(D)$ from (2)? (H2 (i))	Still open
If not, what is the miscoverage deviation? (H2 (ii))	Still open
Structural assumptions needed on $\hat{f}, \hat{g}$ / constrained programs for stability (Wenbin H2)	Still open
Application instance: AI/LLM predictions into downstream optimizer; hypothesis test "trust this decision?" (Meeting 3)	Partially solved (vision stated; instance not fixed)
Gap-aware heterogeneous noisy argmax/top-k as first theorem (H1 §8.4)	Still open (proposed)
Modular calculus mapping heterogeneous pipelines to stability costs then optimizing noise for inference width (H1 conclusion)	Still open
Adaptive regret frontier formulation	De facto not active as of M3; no spoken forever-abandon; revive unknown
Quantization risk-profile primary project	Redirected / no longer primary
Sequential CREAM:	Backup only; not adopted; revive unknown

15 Current state of the research

1. **Agenda (L1):** Meeting 3 — formal stability of optimization-induced policies; discontinuity; AI$\rightarrow$optimizer; HT "trust this decision?"

2. **Working math (L2 stand-in):** H2 — post-hoc UQ with optimization-selected hyperparameter; two-part program; five `[Wenbin]:` comments. Not proven = July 2 Overleaf; Parts I/II empty.

3. **Immediate work authorized (L3):** Brainstorm methodologies for Wenbin's July 2 setup (E1); fill Part I then Part II; method *bias* = no recalibration / data-randomize / structured noise (Comment E) — **bias not forever-lock** while (iii) text still says recalibrate.

4. **Inventory (L5):** Keep using H1 as non-controlling live catalog for primitives / Zrnic–Jordan bridges / Woody-mandated literature neighborhood.

5. **Terminology:** "post-hoc inference" (Nick-attested Wenbin suggestion, 12 Jun).

6. **Incomplete:** H2 stubs; Overleaf identity gap; (iii)/W5 clash; $C(\cdot, \alpha)$ vs θ bug; June 11 notes.

16 Missing companion evidence

Called out so the archive is honest about gaps (from E1 concluding section, still unresolved in this corpus):

- Exact July 2 Overleaf problem setup text. H2 (26 Jul) is the best available stand-in — **do not equate** with July 2 Overleaf.

- Two files attached to Wenbin's July 2 email (Nick's Monday presentation materials).

- June 11 contemporaneous notes (alleged optimization-specific stability intuition; phrase "optimization-based selection" is *not* in email bodies).

- Any earlier email connecting algorithmic stability to replacing data splitting in CREME/CREAM, if it exists.

17 How to use these folders

1. Read **this overview** for timeline, layered authority, and council corrections.

2. Read **T1 Meeting 3** for the spoken agenda (discontinuity, PPI, trust).

3. Read **H2** for `[Wenbin]:` comments and the working UQ math — as L2 stand-in, not as proven July 2 Overleaf.

4. Read **E1** for the July 2 methods handoff and provenance limits.

5. Use **H1** as live inventory — do not treat it as the problem definition.

6. Use `papers/` for PDFs cited above (in-scope for evolution; not in `history/`).

18 Council verdicts (26 Jul 2026 swarm)

Primary sources sifted by parallel forensic agents; attacked by critics; settled by chief judge with dissenting red lines. Full reports in `overview/swarm/`.

Absolute (do not reverse without new primary evidence). July 2 Overleaf preferred but missing; H2 $\neq$ proven identical; H2 mid-edit hybrid; five Wenbin comments; (iii) vs Comment E unresolved; June 25 proposal unapproved by email; M3 spoken agenda as above; H1 non-controlling catalog.

What Wenbin wants Nick to do next. Narrow to his UQ example; fill Part I then Part II; pursue (i)–(iii) with method bias toward data-randomize / structured noise / inverse opt (no recalibration); write methodologies into the draft; stay flexible if a more principal setting appears.

19 Final audit (scoped corpus)

Every in-scope document appears (H1, H2, E1, T1).

H1 triplicate collapsed to one file with MD5 note; content not deleted.

Historical / superseded plans retained with status labels.

Incorrect or redirected ideas retained (Ideas A–C).

Current state identified as L1=M3 + L2=H2 stand-in + L3=July 2 auth.

Major idea evolution paths recorded (without collapsing distinct artifacts).

Open problems collected with status; mid-edit clashes flagged.

Out-of-scope material explicitly listed rather than silently omitted.

Overview overclaims corrected by council (no `companion` = `H2`; no flat H2 crown; no bare Abandoned without caveats).

End of overview.